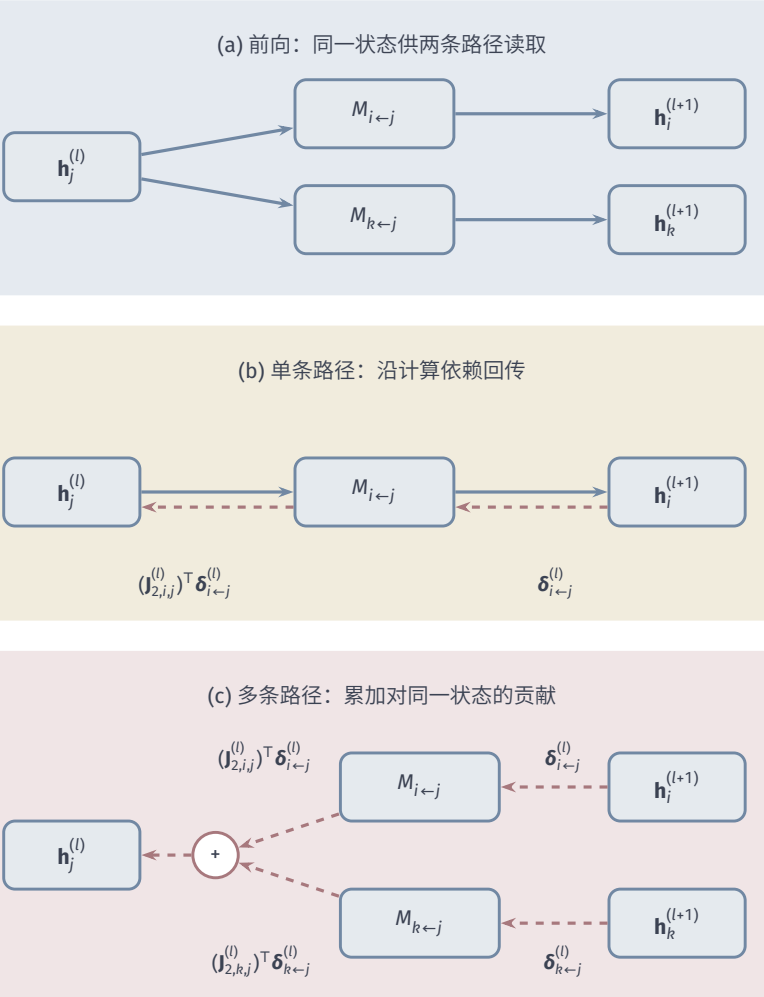

$$\mathbf{g}_{j,\text{send}}^{(l)} = (\mathbf{J}_{2,i,j}^{(l)})^\top \boldsymbol{\delta}_{i\leftarrow j}^{(l)} + (\mathbf{J}_{2,k,j}^{(l)})^\top \boldsymbol{\delta}_{k\leftarrow j}^{(l)}$$